

CALCIUM CARBONATE TABLETS 'MPI' 500MG

NAME AND STRENGTH OF ACTIVE SUBSTANCE:

Each tablet contains:

Precipitated Calcium Carbonate 500mg (equivalent to Calcium 200mg)

PRODUCT DESCRIPTION:

An orange, round of diameter 11mm flat tablet with 'MPI' marking.

PHARMACODYNAMICS:

Calcium is essential for the functional integrity of the nervous, muscular and skeletal systems. It plays a role in normal cardiac function, renal function, respiration, blood coagulation, and cell membrane and capillary permeability. Also calcium helps to regulate the release and storage of neurotransmitters and hormones, the uptake and binding of amino acids, absorption of vitamin B12, and gastrin secretion. The major fraction (99%) of calcium is in the skeletal structure primarily as hydroxyapatite; small amount of calcium carbonate and amorphous calcium phosphates are also present. The calcium of bone is in constant exchange with the calcium of plasma. Since the metabolic functions of calcium are essential for life, when there is a disturbance in the calcium balance because of dietary deficiency or other causes, the store of calcium in bone may be depleted to fill the body's more acute needs. Therefore, on chronic basis, normal mineralisation of bone depends on adequate amounts of total body calcium.

Calcium carbonate has a high neutralisation capacity combined with a rapid rate of reaction make it an ideal antacid management of conditions associated with hyperacidity. It effectively neutralises free acid yet does not causes systemic alkalosis in the presence of normal renal function.

PHARMACOKINETICS:

Calcium carbonate is converted to calcium chloride by gastric acid. Some of the calcium is absorbed from the intestines but about 80% is converted to insoluble calcium salts such as the carbonate and stearate, and excreted in the faeces. Calcium is absorbed from the small intestine. About one-third of ingested calcium is absorbed although this can vary depending upon dietary factors and the state of small intestine; also the absorption is increased during period of high physiological requirement such as during pregnancy and lactation.

After absorption calcium is eventually incorporated into bones and teeth with 99% of the body's calcium content being present in such skeletal tissue. The remaining calcium is present in both the intra-and extracellular fluids. About 50% of the total blood-calcium content is in the physiologically active ionised form with 5% being complexed to citrate, phosphate or other anions and 45% being bound to protein. Excretion of calcium occurs in the urine although a large proportion is reabsorbed in the renal tubules. Excretion also occurs in the faeces, this consisting of unabsorbed calcium as well as that secreted in the bile and pancreatic juice. Minor amounts are lost in the sweat. Calcium cross the placenta and is also excreted in breast milk.

INDICATION:

It is used as an antacid and as a phosphate binder in patients with renal failure, and as a source of calcium for the treatment or prevention of calcium deficiency. Conditions that may be associated with calcium deficiency include hypoparathyroidism, achlorhydria, chronic diarrhoea, vitamin D deficiency, steterrhoea, sprue, pregnancy and lactation, menopause, pancreatitis, renal failure, alkalosis and hyperphosphatemia.

RECOMMENDED DOSAGE:

To be taken orally, with a full glass of water.

Adults:

Hypocalcemia : 1.25g (2 ½ tablet, i.e. equivalent of 500mg of calcium) two to four times a day after meals.

Hyperphosphatemia: 5-13g (10-26 tablets, i.e. equivalent of 2 to 5.2g of calcium) per day in divided doses with meals.

Antacid: 500mg (1 tablet) three to four times a day after meals.

ROUTE OF ADMINISTRATION:

To be taken orally.

CONTRAINDICATIONS:

It is contraindicated in patients with ventricular fibrillation or hypercalcemia.

WARNINGS AND PRECAUTIONS:

1. The risk of acid rebound from high doses or prolonged use should be borne in mind when calcium carbonate is being used as an antacid.

2. Calcium salts should be given cautiously to patients with impaired renal function, cardiac disease, or sarcoidosis.

INTERACTIONS WITH OTHER MEDICAMENTS:

1. Calcium enhances the effects of digital glycosides on the heart and may precipitate digitalis intoxication.
2. Calcium salts reduce the absorption of certain drugs including tetracyclines and phenytoin, because of possible formation of nonabsorbable complex. Patients should be advised not to take calcium supplements within 1 to 3 hours taking the medications.
3. Thiazide diuretics may increase the renal reabsorption of calcium, thereby leading to hypercalcemia.
4. Because the rate and/or extent of absorption of other oral medications may vary when used concurrently with calcium supplements, especially calcium carbonate, patients should be advised not to take any other oral medications within 1 to 2 hours of calcium supplements.

PREGNANCY AND LACTATION:

If you are pregnant or breast-feeding, think you may be pregnant or are planning to have a baby, ask your doctor for advice before taking this medicine.

SIDE EFFECTS:

Calcium carbonate like other calcium salts, may cause gastro-intestinal irritation and constipation. Flatulence from released carbon dioxide may occur in some patients. Side effects may be more likely to occur if calcium supplements are taken in larger doses than recommended (greater than 1-1.5g a day), if taken for a longer period of time, or if taken by patients with renal function impairment. Hypercalcemia after oral administration can occur as can alkalosis following high doses of calcium carbonate, they may resolve on reducing the dose. The milk-alkali syndrome, which includes hypercalcemia and alkalosis together with renal dysfunction, has occasionally occurred, usually in patients taking large doses; patients with renal impairment or dehydration and electrolyte imbalance are predisposed.

SYMPTOMS AND TREATMENT OF OVERDOSE:**Symptoms:**

Gastro-intestinal irritation, constipation and hypercalcemia can occur. Symptoms of hypercalcemia may include anorexia, nausea, vomiting, constipation, abdominal pain, muscle weakness, mental disturbances, polydipsia, polyuria, bone pain, nephrocalcinosis, renal calculi, and, in severe cases cardiac arrhythmias and coma.

Treatment:

A serum calcium concentrations exceeding 10.5mg per 100ml is considered a hypercalcemia condition. Withholding additional administration of calcium and any other medications that may cause hypercalcemia usually resolves mild hypercalcemia in asymptomatic patients, when patient's renal function is adequate.

When serum calcium concentrations are greater than 12mg per 100ml, immediate measures may be required with possible use of the following:

1. Rehydration by either the oral or intravenous route and force diuresis with frusemide or other loop diuretics rapidly increase calcium excretion, is the mainstay of therapy.
2. ECG monitoring and the possible use of beta-adrenergic blocking agents to protect the heart against serious arrhythmias.
3. Possibly including calcitonin and adrenocorticoids in the treatment.
4. Phosphate may be useful, but should be given by mouth and only to patient with low serum-phosphate concentrations and normal renal function. Hemodialysis may be considered as last resort. Careful monitoring of serum-electrolyte concentrations is essential throughout therapy.

DOSAGE FORMS AND PACKAGING AVAILABLE:

Dosage form: Tablet

Pack size : Alu/PVC Blister Pack : 100x 10's

NAME AND ADDRESS OF MANUFACTURER / PRODUCT REGISTRATION HOLDER:

MALAYSIAN PHARMACEUTICAL INDUSTRIES SDN BHD

Plot 14, Lebuhraya Kampung Jawa, 11900 Bayan Lepas,
Pulau Pinang, Malaysia.

DATE OF REVISION:

19/12/2019